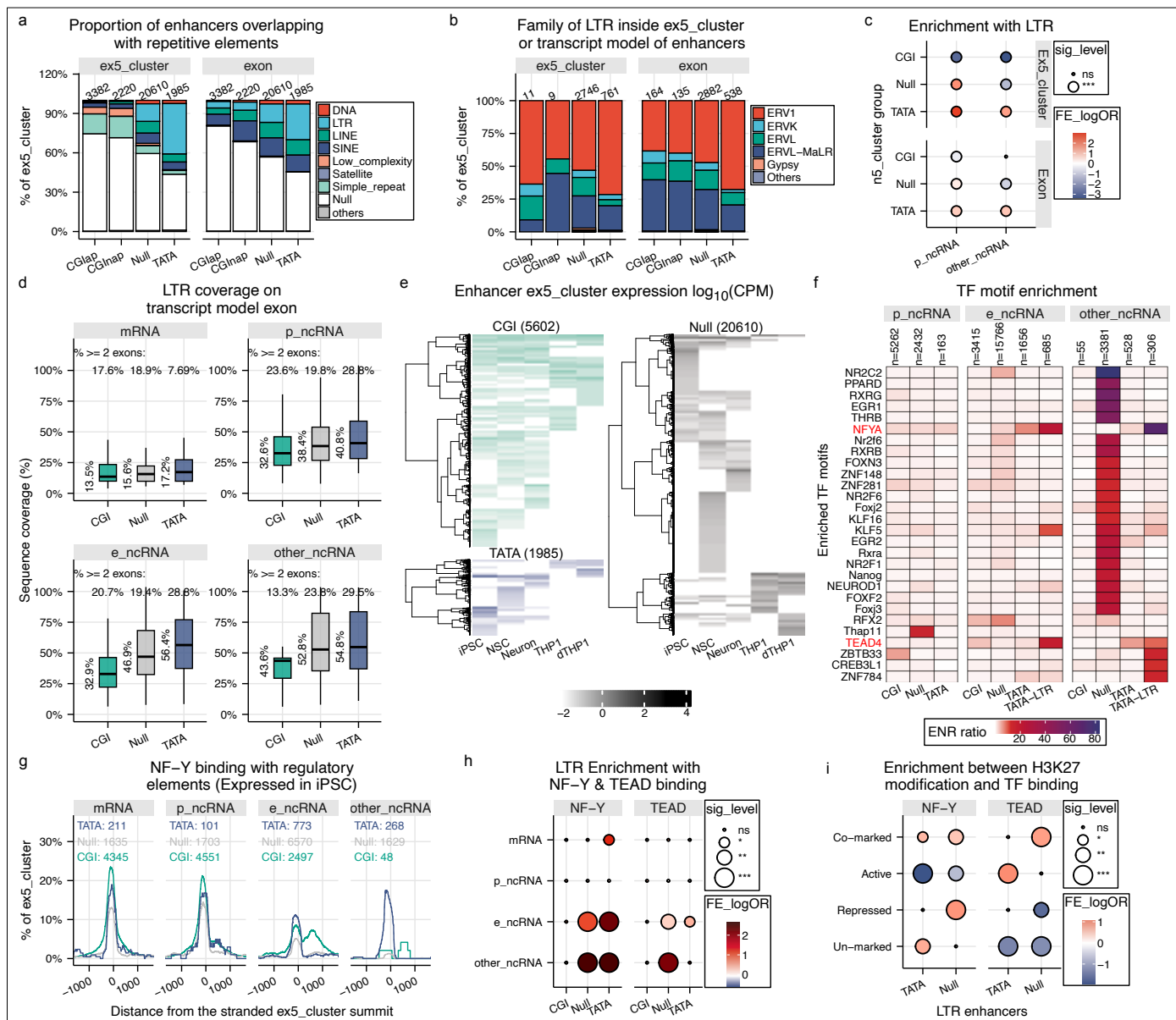




Extended Data Figure 9 | TATA box enhancers enriched with LTR element are activated by NF-Y.

a, Proportion of ex5_clusters and exons (grouped by cluster) intersecting with repetitive elements, applying the same criteria as in **Figure 6e**. **b**, Family composition of LTR elements intersecting enhancers and eRNAs. **c**, Enrichment with LTR across different groups in p_ncRNA and other_ncRNA was tested using Fisher's exact test. **d**, LTR coverage percentage over exons of transcript models that intersected with LTR, grouped by promoter type and regulatory element. Percentages on the top indicate the proportion of transcript models with LTR overlap across ≥ 2 exons. **e**, Expression profile of e_ncRNA ex5_clusters across five cell types, grouped into CGI, Null, and TATA categories. **f**, Enrichment ratio of transcription factor motifs was shown for those with FDR < 0.05 and enrichment ratio > 12 and expression levels > 1 CPM. Motif analysis was performed on non-redundant 501 nt CRE sequences (summit +100 nt & -400 nt). **g**, Distribution of NF-Y binding across ex5_cluster summits, grouped into mRNA, e_ncRNA, p_ncRNA, and other_ncRNA and sub-grouped by CGI, Null, and TATA. The total number of ex5_clusters in each group is shown. **h**, Enrichment between LTR and binding of NF-Y and TEAD considering only ex5_cluster transcribed in iPSC. Fisher's exact derived odds ratio and p-values of the 12 subgroups were shown. **i**, Enrichment between chromatin states and binding of NF-Y and TEAD. Fisher's exact derived odds ratio and p-values were shown from TATA and Null subgroup of e_ncRNA ex5_clusters transcribed from iPSC, including only LTR⁺. Co-marked (H3K27ac⁺ & H3K27me3⁺), Active (H3K27ac⁺), Repressed (H3K27me3⁺), Un-marked (H3K27ac⁻ & H3K27me3⁻). Unless specified, statistical tests were two-sided Wilcoxon; p < 0.05 (*), p < 0.01 (**), p < 0.001 (***), n.s. = not significant.